

Why the Two Flexoelectric Potential Formulas Look the Same

1 Short Answer

The two potential expressions have the same visible shape because both come from integrating a through-thickness electric field of the form

$$E_2(Y) = \text{constant} + \text{constant} \frac{\cosh(Y/\ell_b)}{\cosh(a/\ell_b)}, \quad a = \frac{H}{2}.$$

After integration, both potentials must have the form

$$\phi(Y) = \text{linear term} + \sinh \text{ boundary-layer term} + \text{constant}.$$

Therefore, the visual form of the final potential is not enough to identify the mechanical assumption used in the derivation. The difference is mainly in the coefficients that define the boundary-layer length and amplitude.

2 Same Shape, Different Voltage Reference

The older expression may be written as

$$\phi_{\text{old}}(Y) = -A_0^{\text{old}}Y + B_0^{\text{old}} \frac{\sinh(Y/\ell_b)}{\cosh(a/\ell_b)} + \phi_0. \quad (1)$$

Here ϕ_0 is arbitrary because electric potential is defined only up to a constant.

The new generalized-traction expression was written with the bottom surface as the voltage reference:

$$\phi_{\text{new}}(-a) = 0.$$

It is

$$\phi_{\text{new}}(Y) = -A_0^{\text{new}}(Y + a) + B_0^{\text{new}} \left[\frac{\sinh(Y/\xi)}{\cosh(a/\xi)} + \tanh(a/\xi) \right]. \quad (2)$$

Expanding Eq. (2) gives

$$\phi_{\text{new}}(Y) = -A_0^{\text{new}}Y + B_0^{\text{new}} \frac{\sinh(Y/\xi)}{\cosh(a/\xi)} + [-A_0^{\text{new}}a + B_0^{\text{new}} \tanh(a/\xi)]. \quad (3)$$

Thus Eq. (2) is the same shape as Eq. (1). The extra $\tanh(a/\xi)$ term only fixes the constant so that $\phi(-a) = 0$. It is not an additional physical field.

3 Where the Shorthand Definitions Differ

The two derivations use the same kind of basis function, but they do not necessarily use the same coefficients. The safest way to compare the formulas is not to start from ℓ_b and ξ . Instead, compare the coefficients from which those lengths are built.

Define

$$C_L = \lambda + 2\mu, \quad C_T = \lambda, \quad \beta = \frac{C_T}{C_L}.$$

Then the bulk part of the electric field gives

$$A_0 = \frac{\kappa}{\epsilon} (\mu_T - \mu_L \beta).$$

This part is common. The difference is in the shorthand used for the boundary-layer part.

Quantity	Older shorthand	New generalized-traction shorthand
Primary coefficient	$H_L^* = C_L \ell^2 - \frac{\epsilon_0 \mu_L^2}{\epsilon(\epsilon - \epsilon_0)}$	$G_L = H_L + \frac{\mu_L^2}{\epsilon}$
Transverse coefficient	$H_T^* = C_T \ell^2 - \frac{\epsilon_0 \mu_L \mu_T}{\epsilon(\epsilon - \epsilon_0)}$	$G_T = H_T + \frac{\mu_L \mu_T}{\epsilon}$
Bare model substitution	$H_L^* = C_L \ell^2,$ $H_T^* = C_T \ell^2 \quad (\epsilon_0 = 0)$	$G_L = C_L \ell^2 + \frac{\mu_L^2}{\epsilon},$ $G_T = C_T \ell^2 + \frac{\mu_L \mu_T}{\epsilon}$

The starred quantities H_L^*, H_T^* are the coefficients used in the older reduced formula. The quantities G_L, G_T are the effective coefficients that appear in the higher-order stress after the electric field is eliminated from the open-circuit condition $D_2 = 0$:

$$G_L = H_L + \frac{\mu_L^2}{\epsilon}, \quad G_T = H_T + \frac{\mu_L \mu_T}{\epsilon}. \quad (4)$$

For the default bare generalized-traction model used by the plotting script,

$$H_L = C_L \ell^2, \quad H_T = C_T \ell^2, \quad (5)$$

so

$$G_L = C_L \ell^2 + \frac{\mu_L^2}{\epsilon}, \quad G_T = C_T \ell^2 + \frac{\mu_L \mu_T}{\epsilon}. \quad (6)$$

Therefore H_L^* and G_L are not just two symbols for the same definition. In the bare case with $\ell = 0$,

$$H_L^* = 0, \quad G_L = \frac{\mu_L^2}{\epsilon}.$$

So the older shorthand gives no real boundary-layer length from H_L^* , while the generalized-traction shorthand still gives a finite length if $\mu_L \neq 0$.

4 How the Different Coefficients Propagate

Once H_L^*, H_T^* or G_L, G_T are chosen, the rest of the shorthand follows. This is where the formulas look deceptively similar:

Quantity	Older expression	New generalized-traction expression
Boundary length	$\ell_b = \sqrt{\frac{H_L^*}{C_L}}$	$\xi = \sqrt{\frac{G_L}{C_L}}$
Boundary mismatch	$\eta_{\text{old}} = \frac{H_T^*}{H_L^*} - \frac{C_T}{C_L}$	$\eta_{\text{new}} = \frac{G_T}{G_L} - \frac{C_T}{C_L}$
Potential amplitude	$B_0^{\text{old}} = \frac{\mu_L \kappa \ell_b}{\epsilon} \eta_{\text{old}}$	$B_0^{\text{new}} = \frac{\mu_L \kappa \xi}{\epsilon} \eta_{\text{new}}$

The structural similarity is the source of the confusion. If a reader sees only this second table, it is tempting to think ℓ_b and ξ are the same. They are the same only if the underlying definitions also give

$$H_L^* = G_L, \quad H_T^* = G_T.$$

That is a special coefficient choice, not a consequence of the potential formula itself.

5 Mechanical Difference

If one imposes the local condition

$$\sigma_{22} = 0$$

pointwise as the whole transverse equilibrium statement, then

$$C_L v - C_T \kappa Y = 0, \quad v(Y) = \beta \kappa Y, \quad v'(Y) = \beta \kappa.$$

Substitution into the open-circuit electric relation gives a constant electric field,

$$E_2 = \frac{\kappa}{\epsilon} (\mu_T - \mu_L \beta),$$

and therefore the potential is only linear in Y . There is no sinh boundary-layer term.

This is why a formula that contains a sinh term should not be interpreted as the pure pointwise $\sigma_{22} = 0$ solution. It already contains a higher-order boundary-layer correction. The issue is subtler: the older reduced expression uses a plane-stress-like transverse closure for the bulk response and then adds a boundary-layer correction through H_L^*, H_T^* . The new derivation instead solves the generalized traction condition directly,

$$\sigma_{22} - S'_{222} = 0, \tag{7}$$

with the higher-order free-surface condition

$$S_{222}(\pm a) = 0. \quad (8)$$

Therefore the local stress σ_{22} is allowed to be nonzero inside the beam. It is balanced by S'_{222} , so the generalized traction is zero.

6 Practical Consequence

The two formulas become exactly the same if their coefficient sets are also the same:

$$\ell_b = \xi, \quad \eta_{\text{old}} = \eta_{\text{new}}, \quad B_0^{\text{old}} = B_0^{\text{new}},$$

and if the voltage constants are chosen consistently. Otherwise they can look nearly identical on paper while producing different numerical curves.

For the current plotting script, this is the main distinction:

old comparison: coefficients based on H_L^*, H_T^* ,

new default comparison: coefficients based on G_L, G_T .

In particular, in the bare generalized-traction model, even when the mechanical length scale $\ell = 0$, the term μ_L^2/ϵ can still make $G_L > 0$, giving a finite effective boundary-layer length.